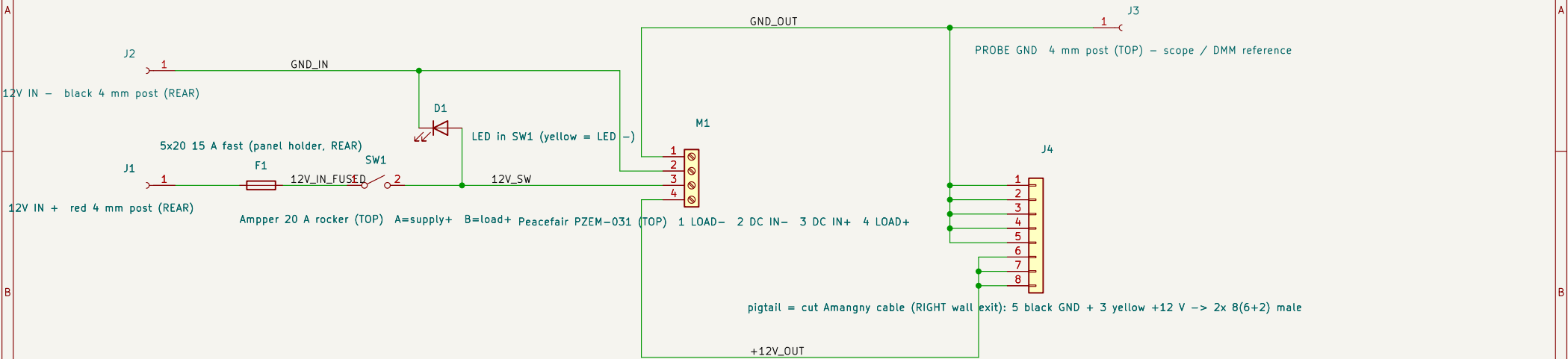


PCIe GPU test rig – power box wiring

Rev B 2026-09-27. Explicit wiring; meter terminals read off the PZEM-031 case label (top->bottom 1..4).



M1 PZEM-031: shunt is internal between 2 (DC IN-) and 1 (LOAD-); + passes straight through 3 -> 4. Fed from the switched side so it sleeps with the rig. 8-100 V, 0-20 A.

SW1 Ampper 3-pin: A = supply +, B = load +, yellow = LED -. LED return goes to the INPUT negative so its current stays out of the reading.

J3 PROBE GND = the OUTPUT negative (LOAD-, after the shunt). Never the input negative: the shunt drop would shift the probe reference under load.

F1 in the + input line ahead of the switch. 15 A fast-blow, never above 20 A (meter / switch limit). Whole path ~240 W: boot stages and rail checks, not a load tester.

Internal bus 12 AWG tinned copper. J4 is the cut Amagncy cable, 18 AWG: 5 black = GND, 3 yellow = +12 V; exits the right wall, zip tie inside as strain relief.

Panels: REAR = J1, J2, F1. TOP deck = SW1, M1, J3 (+ riser dock). RIGHT wall = J4 pigtail exit. FRONT = cord drawer only.

pcie-rig-enclosure/wiring – drawn 2026-09-27, edit in KiCad

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File: pcie_rig_power.kicad_sch

Title: PCIe GPU test rig – power box wiring

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